

Assignment 2

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Title: Perform various data visualization operations on data in Tableau

Aim

To perform and analyse different data visualization operations in **Tableau** using the Sample Superstore dataset, and to build a dashboard that helps understand sales and profit behaviour across segments, regions and product categories.

Dataset

- **Tool:** Tableau Desktop / Tableau Public.
- **Dataset used:** Tableau built-in “**Sample – Superstore**” dataset (or equivalent Superstore CSV/Excel file).
- **Important fields used:**
 - Order Date, Month of Order Date
 - Region – Central, East, South, West
 - Segment – Consumer, Corporate, Home Office
 - Category – Furniture, Office Supplies, Technology
 - Sales, Quantity, Profit

Software

- Tableau Desktop / Tableau Public (latest available version).

Procedure

Step 1: Load data

1. Open Tableau Desktop / Tableau Public.
2. In the **Connect** pane, choose **Sample – Superstore** (or connect to your Superstore data file).
3. After the data source page appears, click **Sheet 1** to start creating visualizations.

Step 2: Create basic visualizations

1. **Pie Chart – Sales by Segment**
 - Drag **Segment** → **Color**.
 - Drag **Sales** → **Angle**.
 - From **Show Me**, select **Pie Chart**.

- Drag **Sales** → **Label** to display values or percentages on slices.
2. **Bar Chart – Sales by Region**
 - Drag **Region** → **Columns**.
 - Drag **Sales** → **Rows**.
 - Choose **Bar** chart from **Show Me**.
 - Click the **Sort** icon to sort regions by descending sales.
 3. **Box Plot – Profit by Category**
 - Drag **Category** → **Columns**.
 - Drag **Profit** → **Rows**.
 - From **Show Me**, choose **Box-and-Whisker Plot**.
 - Optionally drag **Sub-Category** → **Detail** to see more detailed distribution.
 4. **Line Plot – Monthly Sales Trend**
 - Drag **Order Date** → **Columns** and change it to **Month**.
 - Drag **Sales** → **Rows**.
 - From **Show Me**, select **Line**.
 - Drag **Segment or Category** → **Color** if multiple lines are required.
 5. **Area Chart – Monthly Sales by Category**
 - Duplicate the line chart sheet.
 - From **Show Me**, choose **Area** chart.
 - Use **Category** → **Color** to create stacked areas for Furniture, Office Supplies and Technology.
 6. **Histogram – Distribution of Quantity (or Sales bin)**
 - Drag **Quantity** → **Columns**.
 - From **Show Me**, choose **Histogram**.
 - Tableau automatically creates **Quantity (bin)**; bin size can be edited from the data pane if needed.
 7. **Scatter Plot – Sales vs Profit**
 - Drag **Sales** → **Columns**.
 - Drag **Profit** → **Rows**.
 - Drag **Category** → **Color** to differentiate points by category.
 - From **Show Me**, select **Scatter Plot**.

Step 3: Build the dashboard

1. Click **New Dashboard** tab at the bottom.
2. Drag each sheet (Pie, Bar, Box, Line, Area, Histogram, Scatter) onto the dashboard layout as arranged in your screenshot.
3. Adjust sizes and positions so that all visuals are clearly visible with titles and legends.
4. Add filters and interactivity:
 - In one of the charts, drag **Segment or Category** → **Filters** and show filter.
 - Use **Dashboard** → **Actions** → **Add Action** → **Filter** to make a selection in one chart filter the others.

Step 4: Formatting

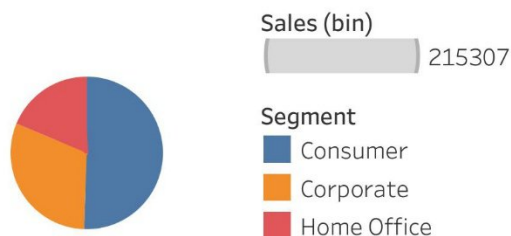
1. Add clear **titles** to each sheet (e.g., “Line Plot – Monthly Sales Trend”, “Histogram – Distribution of Quantity”).
2. Use appropriate **color legends** for Segment and Category.
3. Format axes (labels, tick marks) and tooltips for better readability.
4. Optionally add **Trend Lines** from the Analytics pane to the line or scatter plot to highlight overall trend.

Output

In your manual under **Output**, paste printed screenshots of:

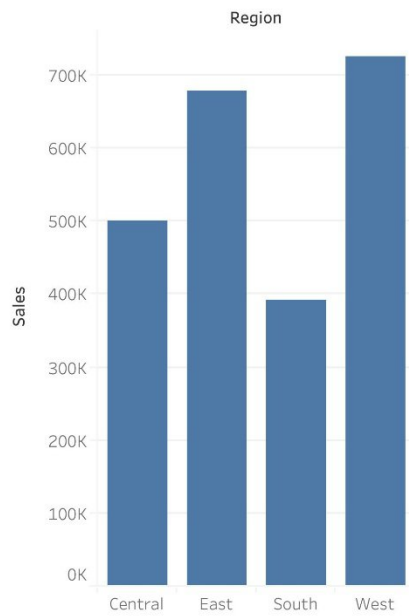
- Pie Chart – Sales by Segment

Pie Chart –
Sales by
Segment



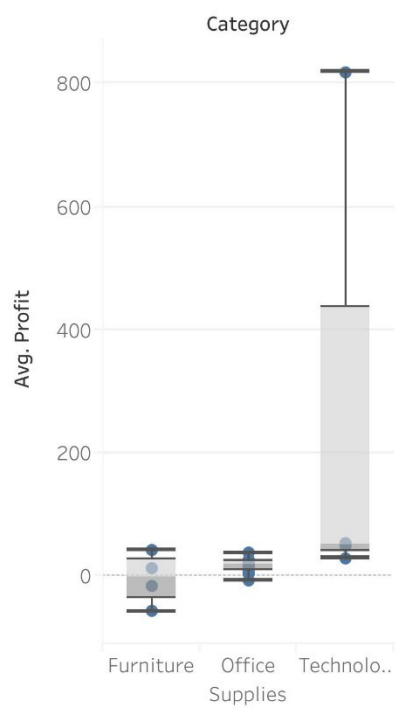
- Bar Chart – Sales by Region

Bar Chart – Sales by Region



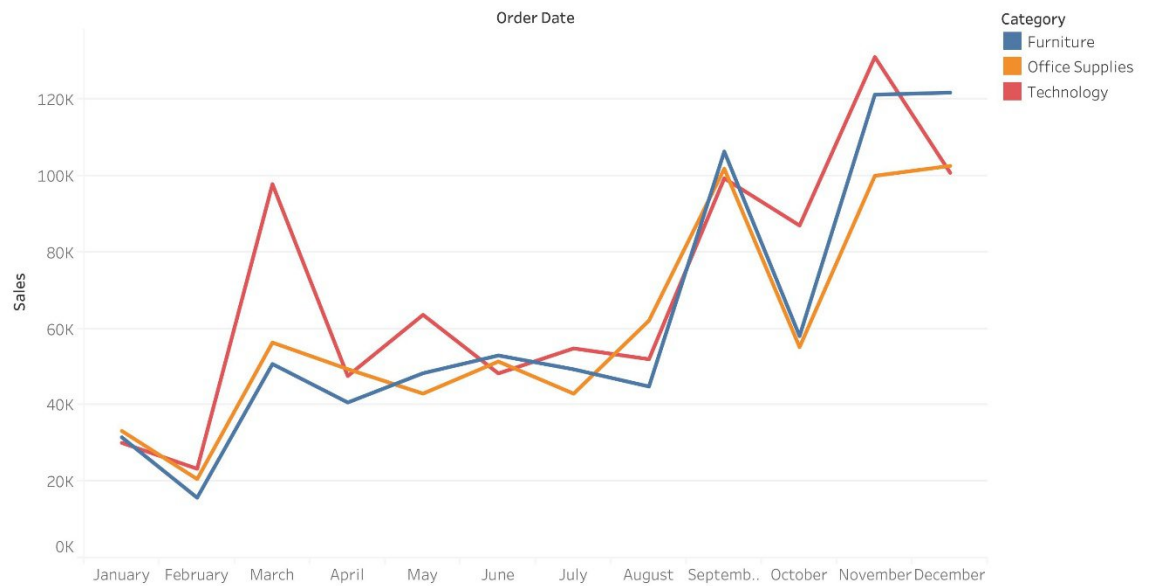
- **Box Plot – Profit by Category**

Box Plot – Profit by Category



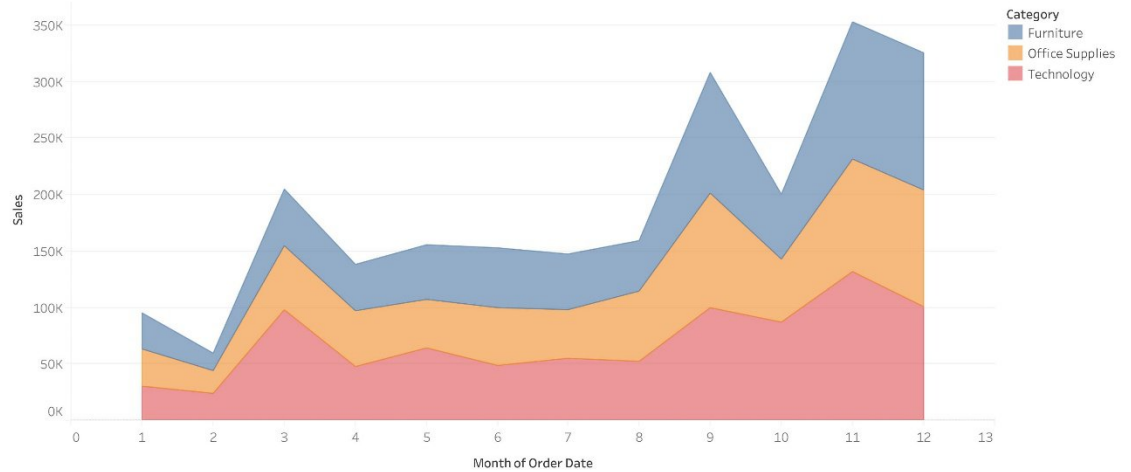
- Line Plot – Monthly Sales Trend

Line Plot – Monthly Sales Trend



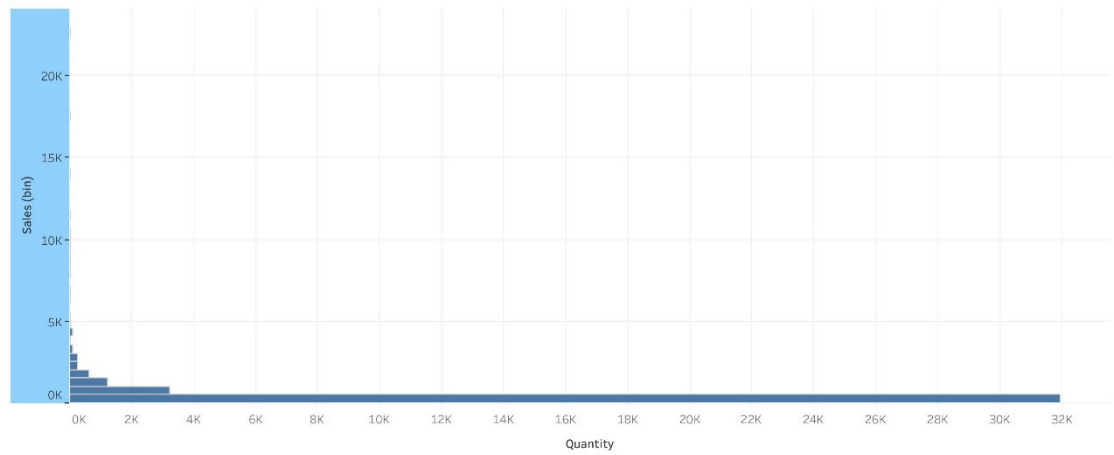
- Area Chart – Monthly Sales by Category

Area Chart – Monthly Sales by Category



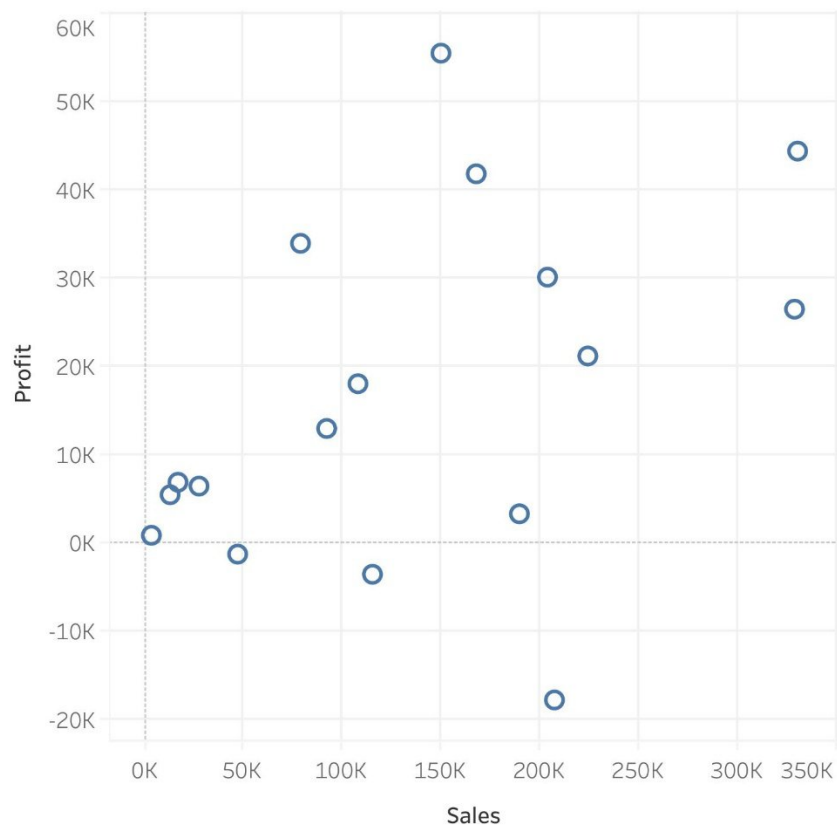
- Histogram – Distribution of Quantity (or Sales bins)

Histogram – Distribution of Quantity

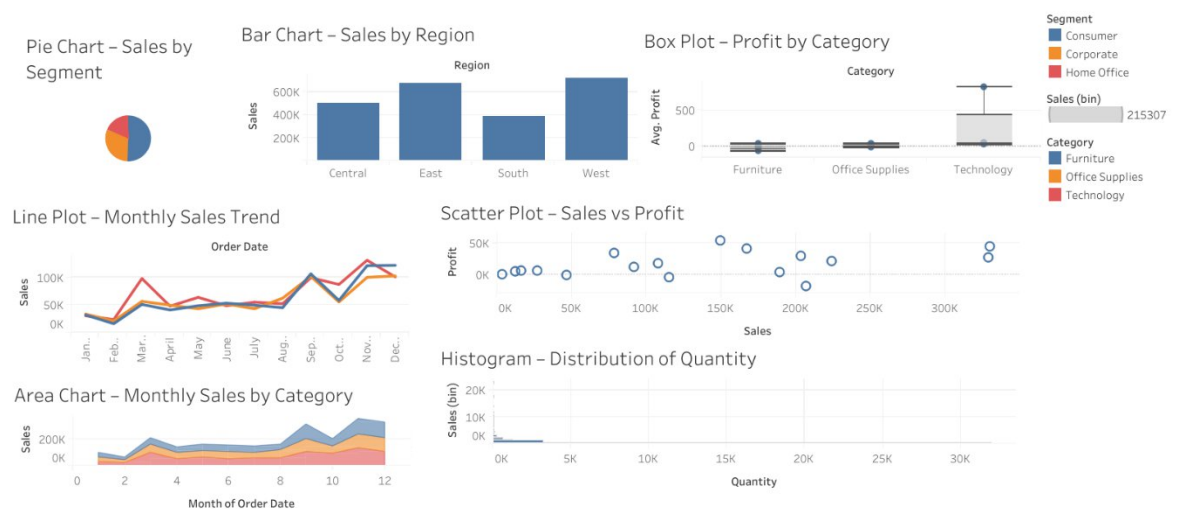


- Scatter Plot – Sales vs Profit

Scatter Plot – Sales vs Profit



- Final Tableau Dashboard containing all the above visuals



Observations

Write points like these:

1. The **line and area charts** show strong seasonal variation in monthly sales and reveal which months and categories generate the highest revenue.
2. The **pie chart** highlights the relative sales contribution of Consumer, Corporate and Home Office segments, indicating which customer segment is most important.
3. The **bar chart for regions** shows that some regions (such as West or East) have significantly higher sales than others, helping to identify strong and weak markets.
4. The **box plot of profit by category** reveals spread and outliers, showing that certain categories (e.g., Technology) have higher median profit but also higher variability and occasional losses.
5. The **histogram of quantity or sales bins** indicates that most orders are of smaller size, with large orders occurring rarely.
6. The **scatter plot (Sales vs Profit)** displays the relationship between revenue and profitability and helps detect high-sales but low-profit orders.
7. The combined **dashboard** allows interactive exploration of the Superstore data, enabling quick comparison of segment, region and category performance in a single view.

Conclusion

Various data visualization operations were successfully implemented in Tableau on the Sample Superstore dataset, including pie, bar, box, line, area, histogram and scatter charts, and an integrated dashboard.

These interactive visualizations make it easier to analyse sales and profit across different dimensions, supporting better decision-making in business analytics and advertising domains.

